

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Geology and Geophysics

2. **Subject Name:** Seismic Methods of Prospecting Lab

L-T-P: 0-0-3

Credits:2

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus: Understanding Seismic Wave Propagation Phenomena through graphical construction of wavefronts for two, three and dipping layer media. Understanding Geometry of Reflection and Refraction/Headwave paths. Insights into Seismic Refraction Prospecting: Numerical Analysis and Synthesis of Refraction/Headwave phenomenon through two, three and multilayer horizontal and dipping interfaces. Modeling and Interpretation of synthetic as well as field Refraction data by Graphical & Analytical methods and Headwave Tomography. Insights into Seismic Reflection Prospecting: Elementary Concepts of Reflection Problems: Raypath as well as Analytical treatment of elementary reflections from horizontal and dipping interfaces in 2-D and 3-D configurations. Raytracing through a multilayer earth model consisting of conflicting dipping horizons. Thinbed stratigraphy and understanding of seismic resolution through convolution of Ricker/Klauder wavelet with reflectivity function. Understanding the geometry of 2-D Seismic Reflection data Acquisition: Construction of CDP diagram and stacking chart for normal and crooked seismic profiles. Estimation of Seismic Velocity: Interval, RMS, NMO, Average and stacking velocity. NMO correction. Determination of velocity and tracing of interface from reflection data. Noise profile analysis and estimation of optimum geophone spacing in a geophone string and planning of seismic field setup for a prospective frontier petroleum lease block. Glimpses of 2-D Seismic data processing. Seismic attribute analysis and elementary interpretation.

Space for entering the syllabus

Reference Books:

- 1) Introduction to Geophysical Prospecting by Milton B. Dobrin and Carl H. Savit
- 2) Applied Geophysics by W. M. Telford, L. P. Geldart and R. E. Sheriff
- 3) Applied Seismology by Mamdouh R. Gadallah and Ray L. Fisher
- 4) Seismic Data Processing Volume I and II by O. Yilmaz
- 5) Introduction to Petroleum Seismology by Luc T. Ikelle and Lasse Amundsen
- 6) Problems in Exploration Seismology and Their Solutions by L. P. Geldart and R. E. Sheriff

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
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1.	Understanding Seismic Wave Propagation Phenomena through graphical construction of wave fronts for two, three and dipping layer media.	06
2.	Understanding Geometry of Reflection and Refraction/Headwave paths. Insights into Seismic Refraction Prospecting: Numerical Analysis and Synthesis of Refraction/Headwave phenomenon through two, three and multilayer horizontal and dipping interfaces.	06
3.	Modeling and Interpretation of synthetic as well as field Refraction data by Graphical & Analytical methods and Headwave Tomography	06
4.	Insights into Seismic Reflection Prospecting: Elementary Concepts of Reflection Problems: Raypath as well as Analytical treatment of elementary reflections from horizontal and dipping interfaces in 2-D and 3-D configurations.	06
5.	Raytracing through a multilayer earth model consisting of conflicting dipping horizons.	03
6.	Thinbed stratigraphy and understanding of seismic resolution through convolution of Ricker/Klauder wavelet with reflectivity function.	03
7.	Understanding the geometry of 2-D Seismic Reflection data Acquisition: Construction of CDP diagram and stacking chart for normal and crooked seismic profiles.	06
8.	Estimation of Seismic Velocity: Interval, RMS, NMO, Average and stacking velocity. NMO correction. Determination of velocity and tracing of interface from reflection data.	06
9.	Noise profile analysis and estimation of optimum geophone spacing in a geophone string and planning of seismic field setup for a prospective frontier petroleum lease block.	03
10.	Glimpses of 2-D Seismic data processing. Seismic attribute analysis and elementary interpretation.	03
Total number of hours		48